

Exploring the Use of Large Immersive Display Systems in the Nuclear Industry

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ABSTRACT

Large immersive display systems play a critical role in the nuclear industry by enabling advanced training, design reviews, scientific visualization, and safety simulations. These systems, such as CAVE and powerwalls, allow engineers and operators to interact with virtual nuclear environments in real-time, providing a deeper understanding of complex systems. Their ability to simulate real-world scenarios for testing and optimization in a safe, controlled environment ensures that operators and engineers can refine process, enhance safety protocols, and troubleshoot complex challenges. This article outlines key lessons learned from deploying these systems, practical insights into their applications, and considerations for future improvements, aiming to guide their broader adoption and effective use in the nuclear industry.

Index terms: Human-centered computing-Visualization

1 INTRODUCTION

The nuclear industry's commitment to safety, precision, and innovation requires the use of tools that enhance operational efficiency and reduce risks. It operates at the intersection of innovation and strict safety requirements, requiring advanced tools that can address both operational and development challenges [1]. Over the years, maintaining a balance between efficiency, precision, and safety has become increasingly complex due to changing technological and regulatory situations. To meet these challenges, the nuclear sector has started to integrate large immersive display systems (LIDS) into its workflows [2]. LIDS have been shown in meeting these objectives by providing dynamic, interactive platforms for simulation, visualization, and training [3].

Unlike traditional tools such as blueprints, static 3D models, or physical mock-ups, immersive display systems enable users to engage in dynamic and interactive environments [4]. These traditional tools, while reliable, often fall short to meet the dynamic and modern requirements and challenges. LIDS enable users to explore complex nuclear systems in detail, improving decision-making and fostering collaboration [5]. These systems transform how engineers, operators, and decision-makers interact with nuclear systems by providing [5]:

- **Comprehensive Visualization:** Users can explore complex reactor designs or facility layouts in high-definition 3D, which fosters a better understanding of system complexities.
- **Interactive Learning Experiences:** Training and simulations allow personnel to practice both routine and emergency procedures in controlled, risk-free environments.
- **Collaborative Platforms:** Interdisciplinary teams can collaborate in real-time, leveraging shared visualizations to optimize designs and decision-making processes.

2 BACKGROUND

Traditional approaches in the nuclear industry, such as physical models and 2D diagrams, have significant limitations when dealing with the complexity and scale of nuclear systems. These conventional tools, while useful, are often challenging to provide an intuitive or complete understanding of multidimensional systems, creating gaps in design validation, operational training, and risk assessment [6]. The increasing need for precision and safety in nuclear operations has driven the adoption of advanced visualization technologies like LIDS, which offer unique solutions to these challenges.

LIDS have emerged as powerful tools that combine high-resolution visualization, interactive simulations, and real-time data integration [7]. These systems allow engineers, operators, and decision-makers to immerse themselves in virtual environments that replicate real-world nuclear facilities and scenarios. LIDS can provide several opportunities for the nuclear industry such as [8]:

- **Bridge Knowledge Gaps:** By presenting detailed, interactive 3D models of reactor designs and facility layouts, immersive systems help stakeholders understand the complexities of nuclear systems without the need for extensive technical expertise.
- **Enhance Training Programs:** Traditional training methods, such as classroom instruction and on-the-job practice, often fall short in preparing nuclear reactor operators for high-risk scenarios. Immersive simulations address this by creating realistic, controlled environments where operators can safely perform both routine and emergency procedures.
- **Validate Designs and Procedures:** Physical prototypes and paper-based reviews can be time-consuming and costly. Immersive systems streamline this process by enabling engineers to virtually navigate designs, identify potential flaws, and test proposed solutions before implementation.
- **Support Collaborative Decision-Making:** The multidimensional nature of immersive environments allows interdisciplinary teams to collaborate effectively, ensuring that all aspects of a project are considered and optimized.

As the nuclear industry continues to evolve, the role of LIDS will expand, providing solutions that enhance safety, efficiency, and

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innovation. By leveraging these technologies, nuclear facilities can address some of their most critical challenges, from maintaining strict safety standards to advancing next-generation reactor designs.

3 TECHNICAL INSIGHTS AND USE CASES OF LIDS

3.1 Transforming Training Scenarios

Immersive systems change the training dimensions from theoretical to experiential. For instance, operators can review reactor shutdowns in environments replicating real-world control rooms, practicing emergency responses to scenarios like leaks or pump failures. These simulations build muscle memory and sharpen critical decision-making skills, ensuring operators are prepared for both routine and high-stakes scenarios.



3.2 Enhancing Design Reviews

Virtual walkthroughs bring clarity to the design process by providing immersive experience to the users. An example is using immersive tools to evaluate the layout of a next-gen reactor system. Engineers can identify potential design flaws, such as inadequate access to critical components, early in the development process. This not only saves time and resources but also enhances safety by optimizing component placement for operational efficiency.



3.3 Preparing for Emergencies

Crisis readiness is non-negotiable, and immersive systems deliver robust preparation. For instance, teams can simulate diverse future events in a nuclear plant, testing the integrity of safety protocols and response plans. By exposing operators to high-pressure, realistic scenarios, immersive training ensures personnel are equipped to handle crises like equipment malfunctions or environmental disasters with confidence and precision.

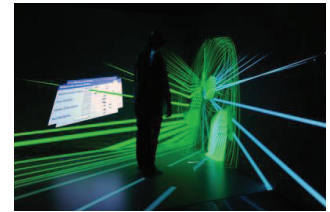
3.4 Engaging Stakeholders for Outreach

These technologies extend beyond operators and engineers. Immersive platforms can be used to educate policymakers and the public by simulating facility tours that highlight key safety measures. For example, stakeholders can virtually explore a reactor containment area, gaining insights into environmental safeguards and operational protocols. This transparency builds trust and fosters informed decision-making.



3.5 Driving Data-Driven Insights

Real-time operational data integrated into immersive environments transforms decision-making. For instance, engineers can visualize live data from radiation sensors overlaid onto a virtual plant layout, enabling proactive responses to anomalies. This predictive maintenance approach not only minimizes downtime but also enhances the overall safety and efficiency of nuclear operations.



4 LESSON LEARNED

One of the most significant lessons learned from deploying LIDS in the nuclear sector is the importance of robust user training and adoption strategies. Many users, particularly those unfamiliar with advanced technologies, initially expressed resistance or doubt. This challenge highlighted the need for comprehensive training programs that combined theoretical instruction with hands-on practice. Demonstrating the practical benefits of immersive systems such as improved training retention and reduced preparation times proved helpful in fostering confidence and acceptance among operators and engineers.

Another critical lesson was the challenge of integrating immersive systems with existing workflows and traditional infrastructure. Aligning new technologies with established tools and processes required collaboration between diverse groups (such as IT) and engineering teams from the beginning. Custom middleware solutions and APIs often became necessary to bridge gaps and ensure seamless data exchange. This experience highlighted the value of early planning and iterative testing to mitigate compatibility issues.

Ensuring realism and accuracy of simulations emerged as another pivotal learning point. Immersive environments must reflect complexities of nuclear systems to be effective. This required leveraging subject matter expertise and incorporating real-world data into scenario development. Iterative feedback from end-users also played a crucial role in refining simulations to make them both realistic and actionable, reinforcing the importance of collaboration throughout the deployment process.

Cost and resource allocation posed a notable challenge as well. Large immersive display systems involve significant upfront investments, not only in hardware but also in software and scenario content development. Ongoing maintenance further added to resource demands. To address this, facilities prioritized high impact use cases where the return on investment was most evident, such as training for high-risk operations or optimizing new reactor designs. Partnerships with academic institutions and industry stakeholders also proved valuable in sharing costs and expertise, enabling broader adoption and implementation.

These lessons collectively highlight the complexity of integrating immersive technologies into the nuclear sector. However, they also demonstrate the transformative potential of these systems when deployed thoughtfully, offering new ways to enhance safety, efficiency, and collaboration.

5 FUTURE DIRECTIONS

5.1 Digital Twin Integration

LIDS and digital twins integration promises interactive models that evolve with real-time data. From predictive maintenance to anomaly detection, the possibilities are amazing.

5.2 Enhanced Accessibility

Affordable, portable systems are the next frontier. By reducing costs and simplifying interfaces, immersive technologies can reach smaller facilities and mobile teams, facilitating easy access.

5.3 Advanced Simulations

Future systems will simulate extreme conditions with unparalleled accuracy. Think high radiation, intense pressure, or severe seismic events—all modeled to test and perfect reactor designs.

5.4 Public Education and Trust

Imagine immersive exhibits that let the public explore nuclear technology up close. By offering transparency and understanding, these tools could reshape perceptions and foster informed dialogue.

6 CONCLUSION

Large immersive display systems are redefining what's possible in the nuclear industry. By addressing hurdles like user adoption, scenario realism, and integration, we can unlock their full potential. The insights and lessons shared here are just the beginning. As these technologies evolve, too will their impact on driving safety, efficiency, and innovation to unique heights

REFERENCES

- [1] J. S. Walker, *Containing the Atom: Nuclear Regulation in a Changing Environment, 1963-1971*. Berkeley, CA: Univ. of California Press, 1992.
- [2] R. Khadka and J. Koudelka, "A 15-Year Retrospective on Immersive Visualization Lessons from the Applied Visualization Laboratory," *PRESENCE: Virtual and Augmented Reality*, pp. 1-15, 2024.
- [3] P. O'Leary, S. Jhaveri, A. Chaudhary, W. Sherman, K. Martin, D. Lonie, E. Whiting, J. Money, and S. McKenzie, "Enhancements to VTK enabling scientific visualization in immersive environments," in *Proc. 2017 IEEE Virtual Reality (VR)*, 2017, pp. 186-194.
- [4] J. Whyte, *Virtual Reality and the Built Environment*. Abingdon, UK: Routledge, 2007.
- [5] R. Khadka and J. Koudelka, "Immersive Visualization Lab at Idaho National Laboratory," in *Proc. 2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 2023, pp. 225-228.
- [6] J. M. O'Hara, J. C. Higgins, W. S. Brown, R. Fink, J. Persensky, P. Lewis, J. Kramer, A. Szabo, and M. A. Boggi, "Human factors considerations with respect to emerging technology in nuclear power plants," *US Nuclear Regulatory Commission*, Washington, DC, USA, 2008.
- [7] E. Lantz, "A survey of large-scale immersive displays," in *Proc. 2007 Workshop on Emerging Displays Technologies: Images and Beyond: The Future of Displays and Interaction*, 2007, pp. 1-es.
- [8] S. Manjrekar, S. Sandilya, D. Bhosale, S. Kanchi, A. Pitkar, and M. Gondhalekar, "CAVE: an emerging immersive technology--a review," in *Proc. 2014 UKSim-AMSS 16th International Conference on Computer Modelling and Simulation*, 2014, pp. 131-136.